JOULE · PHASE D · REAL FOOTAGE

Capture Guide

A complete, on-the-ground guide for the 2026-08-29 shoot: two people, three iPhones (15/17), a possible fourth, no professional camera. Everything you need for the day is in this document — camera placement, exercise order, exact running order, labeling, and what to do if something goes wrong.

MINIMUM SUCCESS FOR THE DAY

One properly calibrated, clap-synced, three-camera squat set. That single clip closes the exact gap this project has listed as "blocked on you" since it began. Everything else in this guide is real bonus value on top of that floor.

Joule Coach · internal working document Sources: CAMERA_PLACEMENT.md · CAPTURE_PROTOCOL.md · STATUS.md · ROADMAP.md

Cheat Sheet

The one page to glance at mid-shoot. Everything here is explained in full further in this guide — this page just gets you unstuck fast.

HARD RULES, NEVER BREAK

- **Feet in frame** on every camera, every rep, whole set.
- **No mirrors** in any frame, ever.
- **Never touch a mounted camera** once calibration starts.
- **Landscape orientation** on every phone, every clip.
- **Clap once**, sharp, in view of all cameras, at the start of every new recording.

CAMERA SPACING (SQUAT RIG)

- Target **60–90°** apart, 3 cameras.
- Avoid any two cameras near **180°** (facing each other).
- Elevate as high as practical — higher beats exactly-2m.
- 4th phone available? Add it to this rig, don't start a second one.

PEOPLE PER FRAME

- **One** for every normal coverage clip.
- **Exactly one** deliberate two-person clip — the crossing/occlusion stress test.

ANGLE BY EXERCISE

- **Bench / overhead press:** true side view. Not optional.
- **Hinge / row:** side view, strongly recommended.
- **Squat, curl, lunge, push-up:** front or 3/4 is fine.

TODAY'S RUNNING ORDER

- `0` Set up + frame-check the squat rig `~20–30 min`
- `1` Checkerboard calibration pass, all 3 cameras `~10 min`
- `2` Fused squat sets, both lifters, cameras fixed `~20–30 min`
- `3` Single-view coverage, other exercises, priority order `rest of session`
- `4` Stress-test clips (crossing, mirror, occlusion, walking) `~10–15 min`
- `5` Copy everything off both phones, back up twice, before leaving

Contents

Eleven sections. Read 1–6 tonight; keep 7–11 with you tomorrow.

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03 Why this exercise order

04 Camera placement, with diagrams

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08 Labeling and file organization

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Tonight, Before You Leave

None of tomorrow works without these. Do them tonight, not at the gym.

- [ ] **Print the checkerboard.** Open `tools/checkerboard_9x6_25mm.html` and print it — **Section 2 walks through exactly how**, and why the print settings matter, step by step.
- [ ] **Mount the printed sheet on something rigid** — cardboard, a clipboard, a laptop lid. A board that bows or wrinkles hurts corner detection more than you'd expect.
- [ ] **Charge every phone to 100%.** Add up the calibration pass plus a full session of exercise clips and you're looking at roughly 45–75 minutes of actual screen-on recording time.
- [ ] **Clear storage.** 1080p at 30fps runs roughly 130–200MB per minute of footage. Budget several free GB per phone.
- [ ] **Bring a way to mount 3 phones rigidly.** Tripods, phone clamps, or anything genuinely stable — a shelf, a stack of plates, gaffer tape to a fixed rack upright. This is not a nice-to-have for the fused station: once calibration starts, a camera that gets bumped silently invalidates everything after it, and the pipeline has no way to detect that it happened. A handheld phone cannot be one of the fused cameras.

> **IF YOU CAN ONLY DO ONE THING TONIGHT**
>
> Print and mount the checkerboard. Everything else on this list is recoverable improvisation at the gym — a missing calibration board is not.

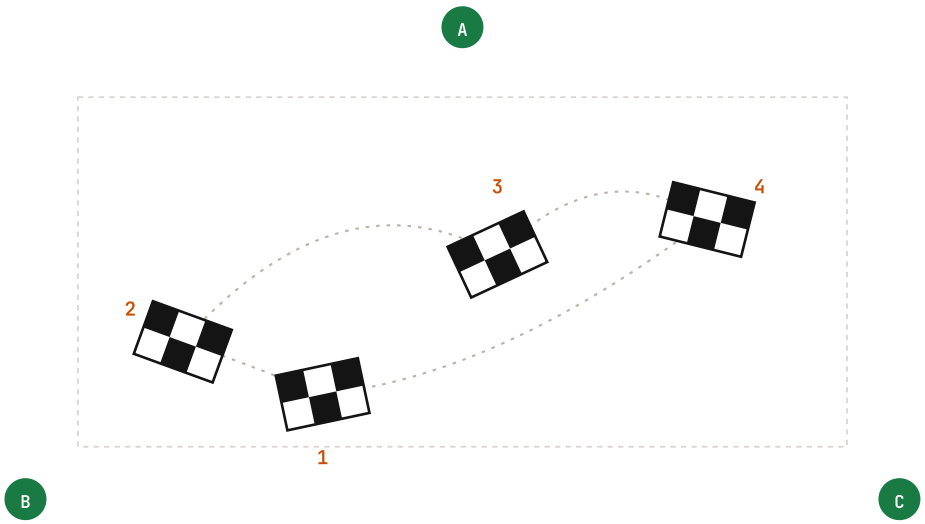
02 · CALIBRATION

How the Checkerboard Actually Works

What it's for, what to physically do with it, and how to print it — in full, not just as a checklist item.

Your phones each see the world flat — a 2D image with no built-in sense of real distance. To combine two or three of them into one accurate 3D measurement, the software needs to work out two things it doesn't know yet: **each camera's own lens quirks** (its exact field of view, and how much it subtly bends straight lines near the frame's edges — every lens does this a little differently), and **exactly where the cameras sit relative to each other** as real geometry, not just an estimated angle.

A checkerboard solves both because it's the one object in the room with **known, exact geometry** — every square is precisely the same size. Shown to the cameras from many different distances and angles, the software can work backward from how that known pattern got distorted and positioned in each frame to solve for both unknowns at once. A single photo of the board can't do this — there isn't enough variation in one view to solve for it. **That's the entire reason you move it around instead of holding it still.**



What "moving it around" looks like from above. The board visits several different spots inside the space all three cameras share (the dashed area), tilted differently each time — not held flat and static, and not just slid sideways without rotating it too. Four to six such positions over about a minute, held steady enough at each one to stay sharp, is what the calibration math actually needs.

WHAT TO PHYSICALLY DO

1. Cameras are already mounted and recording (Phase 1 in the running order, Section 7).
2. Stand roughly where the lifter will stand, holding the board flat, facing generally toward the cameras.
3. Over about 45–60 seconds, move it through several different positions **and** tilt it at a few of them — closer, farther, left, right, then angled up, down, or sideways. Angle variety matters as much as position variety.
4. **Move slowly.** A blurred board can't be detected in a frame at all — a few clean, sharp positions beat many blurred ones.

5. It doesn't need to look tidy on camera. The only goal is genuine variety, visible clearly, across the volume where lifting will actually happen.

> **HOW YOU'LL KNOW IT WORKED**
>
> `calibrate.py` tells you directly when it runs — it prints how many frames it detected the board in per camera, and needs **at least 10 frames where two cameras saw it at the same moment**. If it stops with something like "only 4 frames seen by both cameras," that's not mysterious: the pass was too fast, too short, or spent too long somewhere only one camera could see it. The fix is just to record it again, a little slower, staying in the space every camera shares.

PRINTING IT CORRECTLY

The physical square size is a direct input to the calibration math (`--square-mm 25`) — it's the ruler every real-world distance gets measured against. If a printer silently rescales the page, the software is handed a wrong ruler, and everything measured from that calibration comes out scaled wrong with it.

1. Open `tools/checkerboard_9x6_25mm.html` in a browser, press **Ctrl+P** (Cmd+P on Mac).
2. **Orientation: Landscape.** The board is wider than it is tall.
3. **Paper size: A4.**
4. Find the scale/fit dropdown in the print dialog and set it to **"100%," "Actual size," or "None."** Never leave it on "Fit to page" or "Shrink to fit" — that's the default in most print dialogs, and it's exactly the setting that breaks this.
5. Print, then **measure one printed square with a ruler**, edge to edge. If it doesn't read 25mm, that's fine — write down whatever it actually measured. That real number, not 25, is the one that goes into `--square-mm` tomorrow.
6. Mount it on something stiff — cardboard, a clipboard — so it can't bow.

Why This Exercise Order

Ranked against what the engine actually gets wrong today, not a generic exercise list.

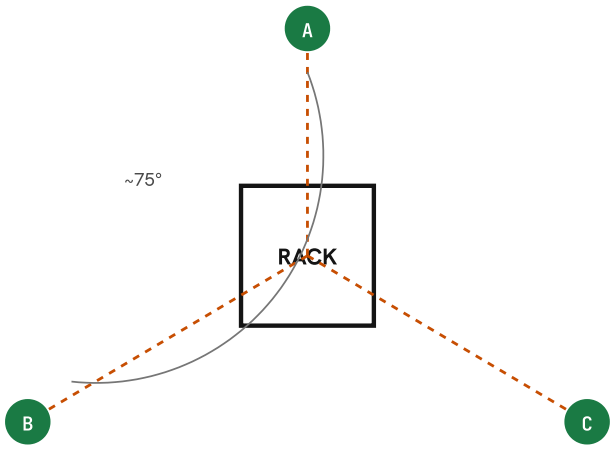
MOVEMENT	REAL STATE TODAY	PRIORITY	WHY REAL FOOTAGE TOMORROW MATTERS
Hinge / deadlift	Not recognized at all. A heuristic was tried and explicitly rejected — it shares a signature with sit-ups.	HIGHEST	Only real labeled data moves this. Not a code fix waiting to happen — a data gap.
Row	Unrecognized, roughly 8% recall.	HIGHEST	Same reason as hinge.
Overhead press / lateral raise	A trained model exists but was mapped to the wrong physical movement during training (a confirmed domain-shift bug).	HIGH	A correctly-labeled real clip directly targets this specific, already-diagnosed bug.
Bench press	Classification works fine; rep *counting* is confirmed weak filmed head-on.	HIGH	The one exercise with an already-confirmed camera-angle fix — side view, not a suggestion.
Reverse lunge	Early support via a hip-ROM heuristic, 6 of 8 correct on validation data.	MEDIUM	More real data solidifies a heuristic that already half-works.
Bicep curl	Early support, 41% recall.	MEDIUM	Same — more data helps an already-working fallback.
Squat	Fully tracked, wider error band, validated only on clean studio data so far.	FLAGSHIP	The real gym floor — equipment clutter, real lighting — is the untested case. Gets full 3D fusion treatment.
Push-up	Tracked, 88% recall on public benchmark data.	MEDIUM	The other exercise studio validation found genuinely hard. Worth a real-floor check.
Sit-up	Classifier recognizes it; the rep counter deliberately counts zero — not built yet.	LOW	A code gap today, not a data gap. One token clip is still useful for whenever the counter exists.

Squat is the one exercise that gets the full three-camera fusion treatment. Every other exercise above gets single-view 2D coverage. A second fully fused station means a second checkerboard pass and a second fixed rig — not realistic for two people alongside everything else in one day. That is a deliberate scope cut, made in advance, not something to feel behind on if you don't get to it.

> **IF A FOURTH PHONE SHOWS UP**
>
> Put it on the squat rig, not a second fused station. Three cameras genuinely beats two for the identity-consistency math this pipeline relies on — a spare fourth phone is worth more there than anywhere else. If you'd rather free it up, it becomes the single-view camera for Section 6's Phase 3, so the mounted rig never has to come down early.

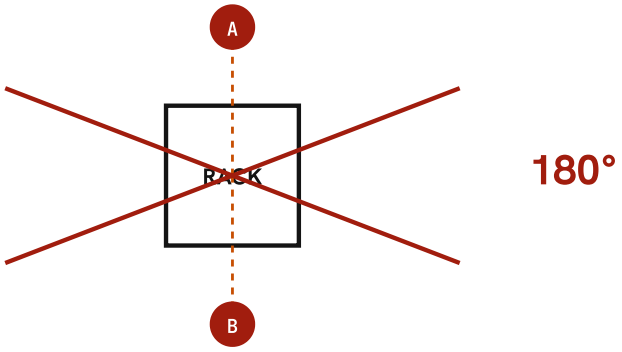
Camera Placement

Three diagrams. These are the geometry rules that actually change what the footage is worth — not generic filming advice.

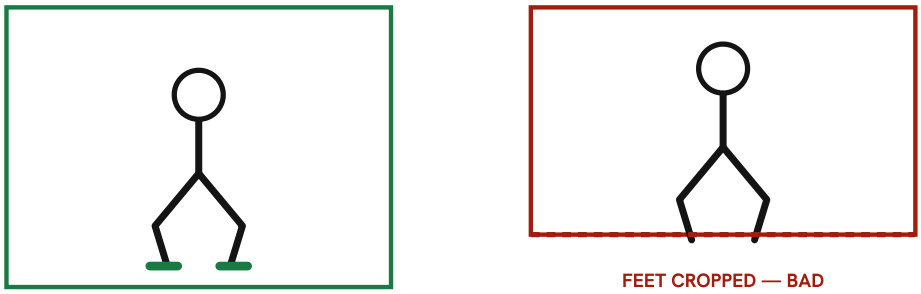


GOOD — 60–90° apart, none opposite

Do this. Three cameras (A, B, C) spread around the rack at roughly 60–90° from each other. This is the spacing that kept both the tightest and the least-favorable studio rigs this project has tested inside a usable accuracy range. Exact degrees don't matter — getting into that rough range does.



Avoid this. Two cameras facing each other directly across the rack. Measured directly on a real multi-camera rig: this exact geometry roughly doubled the joint-angle error compared to a good spacing, on both squat and push-up. If you're down to only two cameras for any reason, this is the one layout to actively steer away from.



Frame wide and high enough that a full squat never pushes a foot out of shot at the bottom of the rep, on every camera. This degrades *silently* — no error, no warning, just quietly worse tracking from that point on. Check it live on-screen before you start recording real sets, not by reviewing footage afterward.

How Many People Per Frame

> **ONE, FOR EVERYTHING EXCEPT ONE CLIP**
>
> Clean single-person footage is what the exercise classifier and the rep counter actually train on. A second person in frame during a normal coverage clip is a confound for that goal, not a bonus — it makes the clip less useful, not more.

The one exception is deliberate: exactly **one** clip with both of you in frame together, crossing or briefly occluding each other's paths. That's a specific stress test for the identity-tracking logic — a completely different system than the exercise classifier, and one that clean single-person clips can never exercise at all. Shoot it once, label it clearly (see Section 7), and move on. More than one such clip adds very little; it's already tested the thing it's meant to test after the first take.

Phone Settings

Set once per phone before you start, tonight if you can.

- [ ] **Landscape orientation.** Every placement rule in this guide assumes it, and nothing in the pipeline has been tested against portrait footage.
- [ ] **Standard Video mode** — not Cinematic, not Portrait/Spatial. Cinematic's synthetic depth blur is a confound with no upside here.
- [ ] **1080p, default frame rate.** No need to reach for 4K or a high frame rate — larger files complicate transfer for no clear accuracy benefit at gym-floor distances.
- [ ] **Lock exposure/focus if your phone offers it** (often called "Lock Camera" or AE/AF lock). Stops the camera hunting exposure mid-set, which otherwise shows up as brightness flicker across a clip.

Running Order

Six phases, in order. Timings are a guide, not a script — the phase order matters much more than hitting a clock.

PHASE 0 Set up the squat rig ~20–30 min

1. Mount 3 phones around the squat rack, roughly 60–90° apart (Section 3). If the space won't allow that spacing exactly, the one thing to actively avoid is any two of the three sitting close to directly opposite each other.
2. Elevate them as much as is practical — higher is the right direction to push toward, even if you can't hit any specific number.
3. Frame check on all 3 phones before recording anything real: stand at the rack, squat once, confirm feet stay in every phone's frame at the bottom of the rep.
4. Check for mirrors anywhere in any of the 3 frames. If the rack is near a mirror wall, reangle rather than partially framing it out.

PHASE 1 Calibration pass ~10 min

1. Start all 3 phones recording as close to simultaneously as you can — a loud "3, 2, 1, go" is enough.
2. One person holds the printed checkerboard in the space where lifting will happen and moves it around for 45–60 seconds: near, far, left, right, tilted. The goal is variety of position, not holding it still — Section 2 explains why and shows exactly what this should look like. All 3 cameras need to see it fully for a good chunk of that time.
3. Stop recording. From this point, do not move, bump, or re-aim any of the 3 phones for the rest of this fused block.
4. Copy the three clips off the phones immediately. Name them clearly: `calib_camA.mov`, `calib_camB.mov`, `calib_camC.mov`.

PHASE 2 Fused squat sets ~20–30 min

1. For each lifter: a clean, correct-form set *first*, before any fatigue. This doubles as the reference footage a real trainer-comparison feature needs — the only version of that feature tested so far compared a clip against itself, so a genuinely separate clean set is real new value.
2. Then a normal working set at real working weight.
3. If either of you has a set that's genuinely slow and heavy (5+ seconds a rep), that's a free stress-test tick — no separate clip needed, just note which take it was.
4. Start/stop recording once per lifter, not per rep. **Clap once, sharply, in view of all 3 cameras, at the very start of every new recording** — a clap from an earlier take does not cover a new one.
5. Copy and label immediately: `squat_<name>_clean.mov` / `squat_<name>_working.mov`, times 3 cameras each.

PHASE 3 Single-view coverage

Remaining time

1. Reposition one phone at a time (the mounted ones can come down now, or use a spare 4th) and work down the priority table in Section 2.
2. Frame check for feet on every new setup, same as Phase 0.
3. Bench press and overhead press/lateral raise need a real side-on view for the pressing arm — bench's is a confirmed requirement, overhead's is the same logic applied as a precaution.
4. Hinge/deadlift and row: side view is the strong recommendation, for the same foreshortening reason, though this specific case hasn't been measured yet the way bench has.
5. One clean clip per exercise is enough. A second take is a bonus, not a requirement.
6. Label each clip with the exercise name as you shoot it.

PHASE 4 Stress-test clips

~10–15 min

1. Both of you crossing/occluding paths — the one deliberate two-person clip.
2. A mirror somewhere in frame, once, deliberately, clearly labeled so it's never mistaken for a real station setup.
3. One deliberate occlusion — step fully behind a machine or pillar mid-set, then continue.
4. One walking or treadmill clip — a cheap regression check.

PHASE 5 Wrap-up

Before leaving

1. Copy every clip off both phones before you leave the gym — see Section 7 for the folder layout.
2. Back up in two places before deleting anything off either phone — laptop plus cloud, or two independent copies at minimum.

Labeling & File Organization

Label as you shoot, not afterward — memory of which take was which fades fast.

```
captures/2026-08-29/
  calibration/    calib_camA.mov, calib_camB.mov, calib_camC.mov
  squat/          squat_<name>_clean_camA.mov, squat_<name>_working_camA.mov, ...
  hinge/          hinge_<name>_camA.mov
  row/  ohp/  lateral_raise/  bench/  lunge/  curl/  pushup/  situp/
  stress/         occlusion.mov, mirror.mov, crossing.mov, walking.mov
```

Kept separate from this project's own `training_data/<class>/` folder on purpose — that folder is pipeline *output* (JSON files the pipeline writes after processing a clip), not a place for raw video. The layout above is just a raw-footage holding area; it gets fed through the pipeline afterward, not stored in `training_data/` directly.

> **ONE CONVENTION, FOLLOWED EVERYWHERE ABOVE**
>
> Exercise name first, lifter name second, camera letter last. It reads in the order you'll actually search for a clip later: "I want the squat footage" before "I want Camera B's footage."

Stress-Test Clips, In Full

Four short, deliberate clips, each testing something specific this project has already found or built — not generic filming advice. One of each is enough.

- [ ] **Two people crossing or occluding paths.** Tests the identity-tracking logic under real clothing, real lighting, real movement — every existing test of this used either a public test clip or a synthetic engineered case, never this project's own camera and gym.
- [ ] **A mirror somewhere in frame, once, deliberately.** The mirror-handling logic has a written, reasoned analysis behind it but has never actually seen a real mirror clip. Include one, on purpose, exactly once, clearly labeled so it's never confused with a real station setup left running.
- [ ] **One slow, heavy set** (5+ seconds per rep — a heavy barbell lift, not a fast bodyweight movement). Already covered if either of your Phase 2 working sets is genuinely heavy and slow — no separate clip needed, just note which take it was.
- [ ] **One deliberate occlusion** — step fully behind a machine or pillar mid-set for a few seconds, then continue. Tests real tracking loss and recovery, not a simulated blackout.
- [ ] **One walking or treadmill clip.** A cheap regression check for logic that already filters out walking from being miscounted as squat reps — confirming it still holds after everything else that's changed since it was last tested.

Troubleshooting

Realistic things that could go sideways tomorrow, and what to actually do about each one.

Q The checkerboard doesn't scan flat, or a corner is bent

> Detection tolerates a small amount of imperfection, but a visibly warped board is worth fixing on the spot — press it flatter against whatever you mounted it on, or re-tape a curling edge, before starting the calibration pass. Cheaper to fix for 30 seconds now than to discover a failed calibration afterward with no way to reshoot.

Q A phone is running low on storage mid-session

> Copy off what you have immediately (Phase 5's folder layout, done early rather than only at the end), delete anything you've already backed up in two places, and keep going. Never delete unbacked footage to make room — if it's a choice between stopping early and losing an already-shot clip, stop early.

Q You forgot to clap at the start of a take

> That take can't be fused in 3D without a manual sync step later (lining up frames by eye against a shared visual event, which is slower and less reliable than a clap). If it's a squat take, just redo it with a clap — cheap insurance given how much the fused squat clip matters. If it's a single-view clip, it doesn't matter at all; single-view footage doesn't need cross-camera sync.

Q Someone bumps a mounted camera after calibration started

> Treat that calibration pass as void. Recording exercise clips on a camera whose exact position calibration assumed is no longer accurate will quietly produce wrong 3D angles with no error or warning — worse than useless, actively misleading. Reset the rig, re-shoot the checkerboard pass, and only then continue. If time is genuinely too short to redo it, downgrade that camera's footage to single-view 2D coverage rather than fusing it.

Q The gym is busy and you can't get a clean single-person shot

> Wait for a gap rather than filming through a crowd — a clip with strangers wandering through frame is a worse version of the deliberate two-person stress test, without being useful as one (it's not labeled or intentional, so it can't be used that way either). Quieter hours, or shooting the less space-hungry exercises first while you wait for the rack to clear, are both fine adjustments.

Q There's no mirror anywhere in the gym

> Fine — skip that one stress-test item rather than forcing it. It's one checkbox out of the whole session, explicitly optional if the environment doesn't offer it. Don't go out of your way to find one; the point is testing a real mirror you'd naturally encounter, not manufacturing an artificial test case.

Q You're running out of time and can't finish everything

> Protect Phases 0–2 above everything else — the calibrated, clap-synced squat set is the actual minimum success for the day, stated on the cover of this guide. Section 2's priority table tells you which single-view exercises to cut first if Phase 3 has to shrink: hinge/deadlift and row are the highest-value ones to protect; sit-up is the safest one to drop entirely.

Q Lighting is uneven between the three camera positions

> Not a hard rule the way feet-in-frame is, but worth a glance: if one camera position is visibly darker or a different color temperature than the others (near a window versus under fluorescent light, for instance), the identity-matching logic can read the same person differently lit as a weaker match than it should. Reposition if an easy fix is available; otherwise note it and move on — it's a soft factor, not a blocker.

After the Shoot

Written down so it doesn't need to be re-derived later — nothing here needs to happen tomorrow.

- [ ] **Run the full pipeline against the squat fusion clip first.** Does the identity-tracking logic hold up with real clothing and real occlusion? Does the fusion step correctly reject a mismatched-person case if one gets staged?
- [ ] **Build a real trainer reference signature** from the clean-form clips captured in Phase 2 — the feature exists and has been tested, just never against footage genuinely separate from what it's compared against.
- [ ] **Feed anything the classifier gets wrong, or any low-confidence pose read, back into the training data** for the next classifier pass — this is what makes tomorrow double as training data, not just a validation exercise.
- [ ] **Write up one finding per stress dimension** — identity, occlusion, tempo, mirrors, 3D fusion — pass or fail plus what specifically broke if it didn't pass. "It ran without crashing" is not a finding on its own.
- [ ] **Build the missing 3-camera rig merge step** if the squat footage used all 3 phones — two pairwise calibration runs need to be combined into one rig file by hand right now; worth turning into a real script once, rather than redoing the merge manually every future session.